॥ श्रीगणेशाय नमः ॥

Series: Number Series and Alphabet Series

Number series

Q: 8, 36, 152, 620, 2496, 10004, ?

→ ×4+4, ×4+8, ×4+12, ×4+24 = 40040

Q: 1, 2, 6, __, 31, 56, 92

→ 1, 2, 6, __, 31, 56, 92
1, 4, 25, 36 ← squares
+9, +16
15

Q: 101, 103, 107, 109, 113, __

→ +2, +4, +2, +4, +2
15

OR

101, 103, 107, 109, 113, 119 ← primes next is 127 ← give higher priority to primes

Q: 52, 51, 48, 43, __, 27, 16

→ -1, -3, -5, -7, -9, -11
36

Q: 2, 3, 6, 15, __, 157.5, 630

→ ×1.5, ×2, ×2.5, ×3, ×3.5, ×4
45

Q: $\frac{2}{3}, \frac{3}{5}, \frac{5}{7}, \frac{7}{9}$, __

→ $\frac{9}{11}$ × primes in numerator, odds in den $\frac{11}{11} = 1$

Q: 1, 4, 27, 16, 125, 36, 343, __

→ 1^3 2^2 3^3 4^2 5^3 6^2 7^3 $8^2 = 64$

Q: 9654237, 654237, 65423, 5423, 542, __

→ 42

Q: 345, 218, 127, __, 29, 10

→ 7^3+2 6^3+2 5^3+2 $4^3+2 = 66$ 3^3+2 2^3+2

Q: 15, 33, 104, __, 2124, 12755

→ ×2+3, ×3+5, ×4+7 = 423, ×6+11

Q. 47, 53, 59, ___, 67, 71

→ primes. 61.

Q. 5760, 960, ____, 48, 16, 8

→ $\times\frac{1}{6}$, $\times\frac{1}{5}$, $\times\frac{1}{3}$, $\times\frac{1}{2}$

= 192

Q. 77, 49, 36, 18, ___.

→ 28, 13, 18, x

77, 49, 36, 18

7×7, 4×9, 3×6, 1×8 = 8

Q. 2, 9, 45, ___, 1485.

→ ×4.5, ×5, ×5.5, ×6

= 247.5

Q. $11\frac{1}{9}, 12\frac{1}{2}, 14\frac{2}{7}, 16\frac{2}{3}$, ___

→ $\frac{100}{9}, \frac{25}{2}, \frac{100}{7}, \frac{50}{3}$

$\frac{100}{9}$ $\frac{100}{8}$ $\frac{100}{7}$ $\frac{100}{6}$ $\frac{100}{5} = 20$

Q. 23, 8, 34, 81, 33, ___

→ 2^3, 3^4, $3^3 = 27$

Q. 5, 6, 12, 15, 60, 65, 390, ___, ___.

→ +1 ×2, +3 ×4, +5 ×6, +7 ×8

= 397 = 397·8

Q. 181, 920, 212, 223, 242, 526. ___

→ 181 920 212 223 242 526 2728 272

Q. 90, 180, 12, 50, 100, 200, ___, 3, 50, 4, 25, 2, 6, 30, 3

→

90, 180, 12, 50, 100, 200, ___, 3, 50, 4, 25, 2, 6, 30, 3

3×50 = 150

Q. 2, 8, 16, 128, ___

→ $2^1, 2^3, 2^4, 2^7$,

1, 3, 4, 7, 11 $2^{11} = 2048$

Q. 1, 11, 21, 1211, 111221, 312211, ——

→ 1

11

21 — two 1's

1211 — one 2 one 1

312211

13112221

Wrong # series

Q. 46080, 3840, 384, 48, 24, 2, 1

÷12 ÷10 ÷8 ÷4 ÷2

x

8

Q. 4, 9, 25, 49, 81, 169

→ 2^2 3^2 5^2 7^2 9^2 13^2 ← squares of primes

x

11^2

Q. 5, 27, 61, 122, 213, 340, 509

→ 2^3-3 3^3 4^3-3 5^3-3 6^3-3 7^3-3 8^3-3

$3^3-3=24$

Q. 3, 10, 27, 4, 16, 64, 5, 25, 125

→ 9

Q. 2, 5, 10, 50, 500, 5000

→ $50 \times 500 = 25000$

Q.

x -2

89, 78, 86, 80, 85, 82, 83

+2 +2

87

Alphanumeric series

Q. AD, CG, FK, JP, ___

→ A → C → F → J → O
+2 +3 +4 +5

D → G → K → P → V
+3 +4 +5 +6

OV

Q. ACE, GJI, MQM, ___

→ A G M S
1 7 13 19

C J Q X
3 10 17 24

E I M Q
5 9 13 17

SXQ

Q. KMS, IP8, GS11, EV14, ___

→ 5, 8, 11, 14, 17

K I G E C
11 9 7 5 3

M P S V Y
13 16 19 22 25

CY17

Q. U, O, I, E

→ A (vowels)

Q. J, F, M, A, M, J, J, ___
→ 10 6 13 1 13 10 10

Jan, Feb, March, ... A (August)

Q. S, M, T, ___, T, F, S
→ W (wednesday)

Q. A, B, D, ___, P

→ 1 2 4 8 16
= H

Q. 2Z5, 7Y7, 14X9, 23W11, 34V13, ___

→ Z Y X W V U

2 +5 7 +7 14 +9 23 +11 34 +13 47

5 7 9 11 13 15

47 U 15

Q. 1ABC, A2BC, AB3C, ___

→ ABC4

Q. AB, BA, ABC, ___, ABCD, DCBA

→ CBA

Q. C4X, F9U, I16R, ___

→ 2^2 3^2 4^2 5^2

C F I L

X U R O

L250

Q. D, B, I, C, ___, ___

4 2 9 3 16 4

P D

Q. 7G, 11K, 13M, ___

→ Primes

G K M 17Q

7 11 13

Q. BC, FGH, LMNO, ___

→ 23 678 12131415 20,21,22,23,24 = TUVWX

45 9,10,11 16,17,18,19

Q. B, FH, LMP, ___

→ 2, 68, 12 14 16, 20 22 24 26 = TUXZ

4 10 18

Repeated series

Q. PQR _ PQ _ PP _ R PP _ RP.

→

PQR _ PQ _ PP _ R PP _ RP.

P

R is followed by P

PQR P PQ R PP Q R PP Q RP.

PP is followed by Q.

Q is followed by R

Q. ab _ aa _ bbb _ aaa _ bbba

a) a, b, b, a

→ try substitution.

abaa bbbb baaaa bbba no repeating pattern ✗

the patterns may not be perfect, substitute options to get whatever seems the most appropriate.

b) baab

ab b aa a bbb a aaa b bbb a

1 2 3 3 4 4 some pattern ✓

Q. _ nmmn _ mmnn _ mnnm _

a) nmmn b) mnnm c) nnmm

→ a) nnmmn mmmnn mmnnmn ✗

b) mnmmnn mmnnn mnnm ✗

c) nnmm nnmmnn mmnn ✓